# 228 Improvement of a Web-Based Question Answering System Using WordNet-Upgraded Chat

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## Abstract

The main purpose of a natural language question answering (QA) system is to find an accurate response to a given question. There are cases, however, in which some factors cause complications when the system tries to capture information. These factors are generally the system shortcomings in "understanding" the question and retrieving a *suitable* answer. In such cases, the questioner not only may want to reformulate the question, but (s)he may want to have a "dialog" with the system. This paper introduces our approach to meliorating the QA system's performance through application of a WordNet-upgraded "trivial dialog" or "chat" with the user.
**Keywords:** QA system, human-computer interaction, natural language processing

## 1 Introduction

Computer applications for natural language QA systems have become a powerful paradigm in Artificial Intelligence (AI) extending beyond AI systems to query processing in database systems as well as to many analytical tasks that involve information gathering, correlation and analysis, all aiming to find an accurate answer to a given question. In open domain QA systems the scope of questions is unrestricted, hence most of those systems use large text collection resources from which they attempt to extract relevant answers. The rapidly increasing availability of information on the World Wide Web (WWW) has made the Web an attractive resource for text collection. This availability has made the QA system a compelling framework for finding information that closely matches the user's needs by providing answers instead of retrieving documents. However, after researching on QA systems in open domain, we realized that certain kinds of questions were inefficiently answered despite the fact that we were exploiting a robust database for potential answers. This is because the questions did not bear enough information as to be properly answered.

Several approaches to open domain QA system have been proposed. Predictive annotation technique [1] and pattern-based approach [2] are some examples. These approaches focus on finding possible answers to a user question using just the information that can be extracted from the question itself. However, in many cases, the possible-answer extraction can not be fulfilled *satisfactorily* since the question does not contain sufficient information to find a suitable response. For example, the incomplete question "Who is a diamond producer?" has a wide spectrum of possible answers when the knowledge base is the Web. Hence, the user's needs are hard to match. This paper presents our approach to QA system in which the application not only aims to find a suitable answer to a question but also tries to bring the user-computer interaction closer to human-human interaction (these terms later illustrated in Chat-Manager section) in order to amend the possible answers selection process, as an endeavor to better match the user's needs. However, dealing with an unrestricted user's chat is a difficult task since the same expression can be said in many different ways, as it is the case of "I am happy", "I am glad", "I am pleased", meaning basically the same emotional state. Therefore, we propose a "WordNet-upgraded chat interactive" QA system that gathers precise information from the user regarding the question and takes advantage of WordNet synonym database to better manage the user chat.

## 2 System Outline

We aim to achieve a QA system capable of holding human-like interaction with the user. Fig. 1 shows our system overview. The system processes the user utterance using a

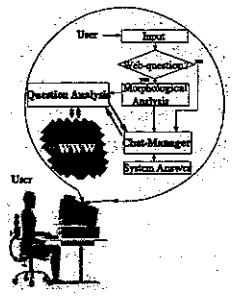



Fig. 1 Chat-interactive QA: System Outline

morphological analyzer. When a Web-question (see section 4) is identified a question analysis process is carried out, and then documents are retrieved using Google API Web search engine. The Chat Manager is in charge of controlling the system interaction with the user. A more detailed explanation of each process is briefly described hereunder.

## 3 Chat-Manager

In order to build a chat-interactive QA system we created a 'chatbot-like ELIZA-clone' with a unique programmed behavior that is in charge of the interaction with the user. ELIZA [3], is one of the most famous examples of Human-Computer Conversation, a computer program that interviews a psychological patient. A chatbot is a computer program that simulates human chatting or trivial dialog. Already in progress is a well known chatbot, 'Alice', openly accessible at www.alicebot.org. In addition, there are sophisticated toolkits as 'Galatea' [4] for building anthropomorphic dialog agent freely available on the Web. Our chatbot is oriented toward obtaining concrete information from the user regarding their request, so as to improve the QA system performance, and to smooth the user-computer interaction.

In addition to the ELIZA-like behavior, the system counts on 180 scripts, including idioms, colloquialisms, etc., in the form of a condition-action rules with a *variable* pattern on the left-hand side (LHS) and its associated set of responses on the right-hand side (RHS), for dealing with the user chat. The following is an example of a variable pattern: [PP/VVx/@NN?/@JJ?/]. Here PP is personal pronoun, VVx is a verb form, @NN is a *variable noun*, @JJ is *variable adjective*, and '?' indicates a part of speech that may or may not be in the match utterance. In this case NN and JJ are variable parts of speech because they are sent to WordNet dictionary [5] database in order to obtain their synonyms. Thus, this pattern can match utterances like "I am *happy* today", "I am *pleased* today", and "I am very *glad*" and use the same pattern-associated set of responses in order to give an answer.

In this way we can handle a wide range of user utterances improving the chattering. We believe that a *chat-interactive* approach to QA system is particularly useful in cases when the user's question lacks sufficient information to elicit a suitable answer, as is the case of many "Who 'is'(or) 'are' [noun]?" questions. Presented below is an example of how the system chat-behavior responds to incomplete user questions:
**System:** Hi! If you have a question I'll search on Internet.
**User:** hello, my name is Joe
**System:** Hello Joe, nice to meet you.
**User:** hi
**System:** Hi! I'll search on Internet for you. *(WordNet-upgraded chat)*
**User:** thanks
**System:** Uhmm, please continue. *(ELIZA-like behavior)*
**User:** yeah...
**System:** What you just said? *(ELIZA-like behavior)*
**User:** who is a diamond producer? *(Web-question detected)*
**System:** I'll search the web. It may take a while and the possible answers may be many. It'll help if you give me more information. What exactly are you looking for? *(Information requested)*
**User:** I'm looking for a diamond producer in America
**System:** So, you said "diamond producer in America"? *(New information obtained) .....*

The question "who is a diamond producer?" when given to any common QA system will retrieve about 305,000 Web pages, with information on diamond producers throughout the world, and since it is a "Who" question, most of QA systems will search for a "person's name" as a possible answer. This overabundance of information makes it very difficult for the system to find the most suitable answer, regardless of how good its question analysis is. However, with a chat-interactive QA system more specific information about what the user is trying to find could be obtained. In the example shown above, the acquisition of the new data " America", reduced the number of retrieved pages by about 48% and provided information about 'Canada' as diamond producer in *North* America within the first several Web pages, which means the chat-interactive QA system can perform a more focused search.

## 4 Question Analysis

This system determines a Web-question (the answer of which is found on the Web) by detecting question words such as "what", "who", "when", "which", and so on. Exceptions include questions about the system itself such as, "what is your name?", and the like. When a Web-question is detected a morphological analysis is carried out. The result of the morphological analysis is needed in order to extract specific part of speech from the question for building a query. We focus on 'who' questions because they tend to lack information (as shown in the chat example in the previous section). Scripts have been specifically designed to deal with them. The query is formed by extracting nouns, adjectives, adverbs and verbs (with some exception, like the verb "to be") from the given question. While the question is being analyzed the system attempts to obtain information from the user by chatting. After the Web-question is detected the following user utterance is morphologically analyzed for extraction of nouns, adjectives, etc., to be added to the previously formed query. The morphological analysis of our question example is: [who(WP)is(VBZ)a(DT)diamond(NN)producer(NN)?(SENT)]. Here the nouns (NN) are extracted to create a keyword query: *diamond producer*. In our example, the new information obtained from the user is "America"(NN); thus, the formed new query becomes *"diamond producer America"*. This new query is sent to the search engine and documents containing possible answers are retrieved. The system analyzes the first 20 HTML Web pages out of the thousands retrieved. Those HTML Web pages are parsed using HTML-Parser and segmented into sentences using LingPipe. From those sentences, the ones which contain keywords from the queries are extracted and ranked. The sentences with more than half the words from the query are considered to be potential possible answers. Some of the sentences extracted for our example are:
–*Canada*: World's Third Largest **Diamond Producer**, Diamonds Net (Rapaport, January 4, 2004) According to ...–
–Of major significance was the opening of North **America**'s first **diamond** mine, the Ekati Mine in North Western Territories in *Canada*.–
–Final feasibility studies for another **diamond producer** in *Canada* are nearly complete...–

## 5 Experiment and Results

The experiment attempted to evaluate how the performance of the QA plain system (without chat) is improved by the addition of the chat-interactive characteristic with WordNet. We selected twenty "Who" questions from a corpora publicly available [6, 7] and compared the possible answers extracted by the chat-interactive system and by the system without Chat-Manager. We observed that the plain QA system could not *satisfactorily* retrieve possible answers to questions like "who is Philip Baity?" because of the huge amount of ambiguous information in the Web. Some of the sentences extracted as possible answers for this question were:
–Philip Baity -all swims best times- ...–
–A.Philip Baity, PA (Alumni) –

On the other hand, when the same question was given to the chat-interactive QA system, new information [law(NN)] was obtained by the system. The possible answers extracted were:
–A.**Philip Baity** (803) 547-2928 -Areas of **law** covered: Highway...–
–... defense lawyer **Philip Baity** says the taping occurred... –

This experiment found that around 26% of the questions selected were better answered by the chat-interactive system which can be considered an improvement from the plain system.

## 6 Conclusion and Future Work

This paper proposes a chat-interactive QA system that uses a WordNet-upgraded ELIZA-like chatbot. The experiment illustrates that the implementation of the chat-interactive QA system affords more accurate possible-answer extraction, improving the performance of a plain QA system, and proving this approach to be effective. Future works will be oriented toward broadening the chatbot semantic database in order to deal with a wider spectrum of questions, a larger variety of user colloquialisms, and finally, toward exploring the benefits of integrating VoiceXML into the Chat-Manager.

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